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From Financial News to Portfolio Risk

Explainable AI/NLP Financial Risk Intelligence Platform

THE QUESTION

Which news should actually change portfolio risk?

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WHAT THE PLATFORM DOES

- Ingests live RSS news and deterministic replay scenarios
- Converts unstructured text into structured risk signals
- Separates sentiment from materiality
- Requires corroboration and policy checks before stress execution
- Feeds one signal into tactical rebalancing and strategic stress testing

DECISION PRINCIPLE

Low-materiality signal visible, not stress-tested	FILTER
Corroborated material event eligible for portfolio stress testing	ACTION

One engine, one structured risk signal

Every article is converted into a machine-readable signal; materiality is judged separately from sentiment.

Problem statement alignment

The platform processes unstructured financial text from multiple sources and produces machine-readable signals for downstream applications.

Negative sentiment ≠ automatically high portfolio risk

Company mention ≠ automatically material exposure

Sentiment score
−1.0 to +1.0, based on FinBERT probabilities

Event classification
One of eight financial event categories

Impact score
Explainable severity score from 1 to 10

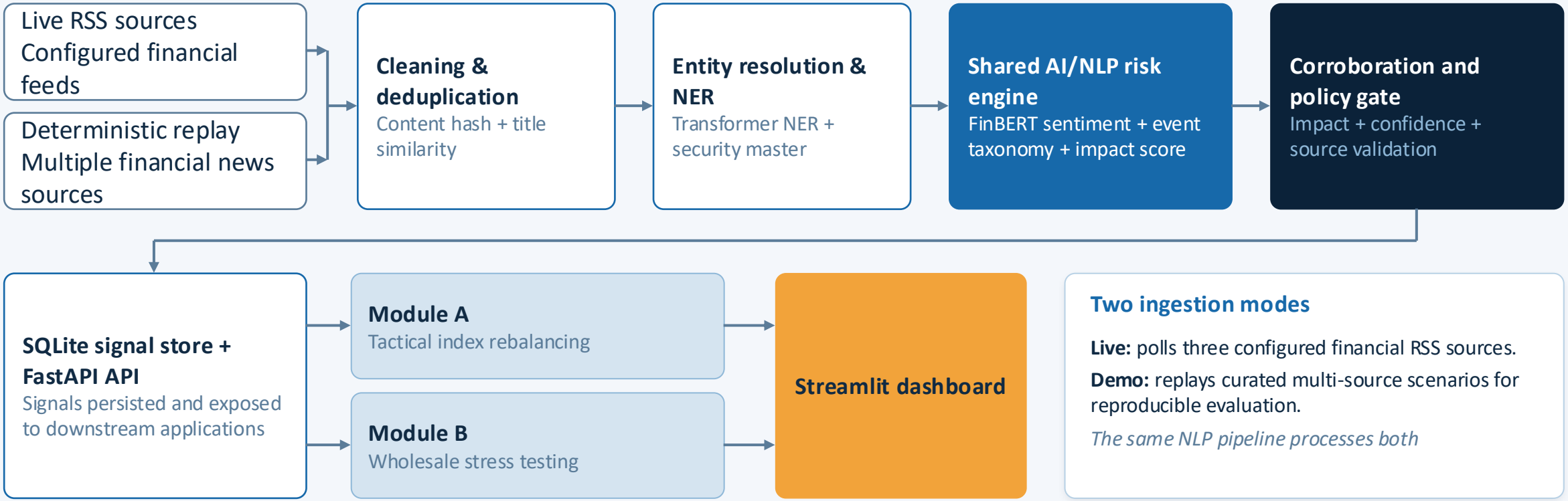
Entity information
Company, ticker, sector and security-master mapping

Evidence and lineage
Source, article text, URL, model version, evidence

Text interpretation → **Materiality assessment** → **Policy decision** → **Portfolio action**

Architecture and data flow

Both ingestion modes run through one shared NLP pipeline and feed two downstream modules.



Two ingestion modes

Live: polls three configured financial RSS sources.

Demo: replays curated multi-source scenarios for reproducible evaluation.

The same NLP pipeline processes both

Built with FinBERT · Transformer NER · SQLite · FastAPI · Streamlit · RSS feeds

Implementation highlights

Explainable NLP, corroboration-aware governance, and multi-asset stress valuation.

Explainable NLP pipeline

- HTML and URL cleaning
- Exact-hash and near-duplicate detection
- Transformer NER with security-master fallback
- FinBERT positive / neutral / negative probabilities
- Hybrid event classifier with evidence text
- Multi-factor impact score: event materiality, sentiment, entity directness, confidence, source reliability

Corroboration-aware governance

≥ 7.0	Impact score
≥ 0.65	Confidence
≥ 2	Independent corroborating sources

Low-materiality or weakly supported signals stay visible for auditability but never trigger a formal stress run.

Multi-asset stress valuation

Equities
Market value and beta sensitivity

Corporate bonds
Duration and credit-spread sensitivity

Loans
EAD, PD and LGD expected-loss change

Swaps
DV01 and interest-rate sensitivity

Illustrative prototype assumptions, not calibrated production risk parameters.

How a Material News Event affects PORTFOLIO RISK

The system converts an eligible material event into an explainable portfolio stress result.

LOW-MATERIALITY SIGNAL

NOT STRESS-TESTED

SIGNAL TYPE	Unconfirmed company-related update
ASSESSMENT	Low-materiality event
EVIDENCE	Single unverified source
OUTCOME	Retained for review; not eligible for formal stress testing

Entity recognition is kept separate from financial materiality.

CORROBORATED MATERIAL EVENT

GATE: ACCEPTED

Multiple independent sources describe the same material event:

Evidence 1 Material event and impact indicators

Evidence 2 Independent confirmation and source validation

Material event · 2 independent sources

ILLUSTRATIVE PORTFOLIO OUTPUT AFTER STRESS-TEST

$192.0M

Portfolio value before stress

−$4.62M

Estimated loss from stress scenario

$187.4M

Portfolio value after stress

PORTFOLIO CHANGE
−2.41%

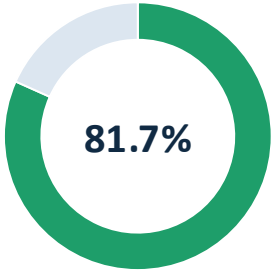
WHERE THE LOSS COMES FROM?

The estimated loss is broken down by asset class, issuer, sector and individual position. Only material, corroborated events receive targeted portfolio shocks.

Evaluation and domain impact

Modest external checks, honestly scoped, and a traceable path from article to portfolio decision.

Financial PhraseBank



FinBERT accuracy

60 balanced sentences
20 positive
20 negative
20 neutral

Small smoke evaluation, not a claim of general or production accuracy.

Versus a sentiment-only approach

SENTIMENT-ONLY

Every negative headline is an alert; every company mention is exposure.

THIS PLATFORM

Only material, corroborated events trigger stress tests; duplicates are collapsed.

US Financial News Articles: unlabeled ingestion test

JSON records → Internal article schema → Cleaning + source validation

No gold sentiment or event labels exist, so no classification accuracy is claimed here.

Domain impact

A traceable path for the risk analyst

Source article
↓
Structured signal
↓
Policy decision
↓
Portfolio impact

Faster triage, fewer duplicate alerts, and more explainable portfolio decisions.

Assumptions, limitations, next steps

A prototype with clear boundaries and a concrete path to production-grade validation.

Prototype assumptions

- Security master, users, replay scenarios and exposures are curated or synthetic
- Stress shocks are illustrative and manually defined
- Event and impact scoring are rule/model hybrids
- SQLite and Streamlit authentication suit demos, not production
- Live RSS depends on network access and publisher feeds

Governance positioning

SR 11-7 | BCBS 239

The workflow is inspired by SR 11-7 model-risk and BCBS 239 data-lineage principles.

Not a claim of Basel compliance or regulatory certification.

Next steps

1. Larger held-out sentiment, event and entity benchmarks
2. Calibrate event-to-market and credit shocks on historical data
3. Historical Module A weight tracking and outcome validation
4. Replace demo authentication and SQLite with production services
5. Scalable streaming and monitoring infrastructure

The prototype demonstrates how unstructured financial information can become structured, corroborated, and explainable portfolio risk intelligence.